dirac16complex in the primordial, modern and future gravitational fields: Euler-Lagrange equations, Kohn-Sham DFT, and the proof that universes of masses {+m, -m} are created in pairs

All statements are computed with exact arithmetic (symbolic or integer) or, where stated, with a convergent numerical method, in the notebook 6planes.nb, Parts 5-7. All equations used are written out here.

1. The field and its Lagrangian in a gravitational field

dirac16complex is the 16-component Pin(4,4) spinor field ψ with complex anticommuting (Grassmann) components, $\bar\psi = \psi^\dagger\sigma_{16}$, $\sigma_{16} = T^0T^1T^2T^3$, where the real Dirac matrices $T^A = (T16^A)[A]$ satisfy $T^AT^B + T^BT^A = 2\eta^{AB}I_{16}$, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$. A gravitational field is a co-frame (vielbein) $e^A{}_\mu$ with inverse $E_A{}^\mu$, metric $g_{\mu\nu} = \eta_{AB}e^A{}_\mu e^B{}_\nu$ and Christoffel symbols $\Gamma^\nu{}_{\mu\lambda}$. The covariant Lagrangian density is

$$\mathcal{L} = \sqrt{|g|}\left[\tfrac{1}{2}\left(\bar\psi\gamma^\mu D_\mu\psi - (D_\mu\bar\psi)\gamma^\mu\psi\right) - m\bar\psi\psi - \frac{\lambda}{2}(\bar\psi\psi)^2\right],$$

$$\gamma^\mu = E_A{}^\mu T^A, \qquad D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi, \qquad D_\mu\bar\psi = \partial_\mu\bar\psi - \bar\psi\Omega_\mu, \qquad \Omega_\mu = \tfrac{1}{4}\,\omega_\mu{}^A{}_B\,T_AT^B, \qquad T_A = \eta_{AB}T^B,$$

$$\omega_\mu{}^A{}_B = e^A{}_\nu\left(\partial_\mu E_B{}^\nu + \Gamma^\nu{}_{\mu\lambda}E_B{}^\lambda\right).$$

The sign of Ω_μ is fixed by covariant constancy of the curved Dirac matrices, $\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\gamma^\lambda + [\Omega_\mu,\gamma^\nu] = 0$, verified exactly in all three fields below.

2. The three gravitational fields

All three are diagonal vielbeins $e = \mathrm{diag}(e_0, e_1, \ldots, e_7)$ of the form used in Pre-gravity6plane.nb.

Primordial field (exactly the vielbein of Pre-gravity6plane.nb with the inflationary choice $a_4(Hx_4) = Hx_4$), $0 < 6Hx_0 < \pi/2$:

$$e_P = \mathrm{diag}\left(\tan 6Hx_0,\ \frac{e^{-Hx_4}}{\sin^{1/6}6Hx_0}\ (\times 3),\ 1,\ \frac{e^{Hx_4}}{\sin^{1/6}6Hx_0}\ (\times 3)\right), \qquad \sqrt{|g|} = \sec 6Hx_0.$$

Modern field (static field of a mass along the hidden direction, Schwarzschild-type profile), $x_0 > r_s$:

$$e_M = \mathrm{diag}\left(f^{-1/2},1,1,1,f^{1/2},1,1,1\right), \qquad f = 1 - \frac{r_s}{x_0}, \qquad \sqrt{|g|} = 1.$$

Future field (mass plus dark energy, Schwarzschild-de Sitter-type profile), between the two horizons:

$$e_F = \mathrm{diag}\left(f^{-1/2},1,1,1,f^{1/2},1,1,1\right), \qquad f = 1 - \frac{r_s}{x_0} - H_\Lambda^2x_0^2, \qquad \sqrt{|g|} = 1.$$

3. Euler-Lagrange equations of dirac16complex in the three fields

Varying ψ_a^* with left Grassmann derivatives, $\frac{\partial^L\mathcal{L}}{\partial\psi_a^*} - \partial_\mu\frac{\partial^L\mathcal{L}}{\partial(\partial_\mu\psi_a^*)} = 0$, gives in every field, exactly,

$$\sqrt{|g|}\,\sigma_{16}\left(\gamma^\mu D_\mu\psi - m\psi - \lambda(\bar\psi\psi)\psi\right) = 0,$$

and the Lagrangian is exactly real, $\mathcal{L}^* = \mathcal{L}$. For diagonal vielbeins $\gamma^\mu\Omega_\mu = \sum_A T^A\frac{1}{e_A}\frac{1}{2}\partial_A\ln\frac{\sqrt{|g|}}{e_A}$ (verified), and only the T^0 term survives. Written out:

Primordial:

$$\cot(6Hx_0)\,T^0\partial_0\psi+e^{Hx_4}\sin^{\frac{1}{6}}(6Hx_0)\sum_{i=1}^{3}T^i\partial_i\psi+T^4\partial_4\psi+e^{-Hx_4}\sin^{\frac{1}{6}}(6Hx_0)\sum_{j=5}^{7}T^j\partial_j\psi-3H\cot^2(6Hx_0)\,T^0\psi=(m+\lambda\bar\psi\psi)\psi.$$

Modern ($f=1-r_s/x_0$) and **future** ($f=1-r_s/x_0-H_\Lambda^2x_0^2$):

$$\sqrt{f}\,T^0\partial_0\psi+\sum_{i=1}^{3}T^i\partial_i\psi+\frac{1}{\sqrt{f}}T^4\partial_4\psi+\sum_{j=5}^{7}T^j\partial_j\psi+\frac{f'(x_0)}{4\sqrt{f}}\,T^0\psi=(m+\lambda\bar\psi\psi)\psi,$$

with $\frac{f'}{4\sqrt{f}}=\frac{r_s}{4\sqrt{x_0^3(x_0-r_s)}}$ (modern) and $\frac{f'}{4\sqrt{f}}=\frac{r_s-2H_\Lambda^2x_0^3}{4\sqrt{x_0^3(x_0-r_s-H_\Lambda^2x_0^3)}}$ (future). For configurations of dirac16complex that depend only on x_0 and x_4, only the T^0 and T^4 terms remain.

4. The DFT-motivated Kohn-Sham approximation

Kohn-Sham (mean-field) treatment of the quartic term. In the spirit of density-functional theory, the interacting field is replaced by non-interacting Kohn-Sham quanta moving in a self-consistent effective potential that depends on a local density:

$$\frac{\lambda}{2}(\bar\psi\psi)^2\longrightarrow\lambda\rho_s\,\bar\psi\psi-\frac{\lambda}{2}\rho_s^2,\qquad\rho_s(x)=\langle\bar\psi\psi\rangle_{KS}=\sum_{n\ \mathrm{occ}}\bar\phi_n(x)\phi_n(x),\qquad M(x)=m+\lambda\rho_s(x).$$

Kohn-Sham equation. For a static field with lapse $B=e_4$ and hidden scale $N=e_0$, stationary modes $\psi=e^{-i\epsilon x_4}u(x_0)$, the rescaling $u=(\sqrt{|g|}/N)^{-1/2}\chi$ (which removes the spin-connection term) and the tortoise coordinate $d\rho=(N/B)\,dx_0$ give

$$h_{KS}\chi=\epsilon\chi,\qquad h_{KS}=\alpha p+W(\rho)\beta,\qquad p=-i\frac{d}{d\rho},\qquad\alpha=-T^4T^0,\qquad\beta=-iT^4,\qquad W=B\,M.$$

α, β are Hermitian, $\alpha^2=\beta^2=I_{16}$, $\alpha\beta+\beta\alpha=0$, and both commute with the Krein operator $K=-i\sigma_{16}T^4$; so h_{KS} is 8 copies (4 with $K=+1$, 4 with $K=-1$) of the 2×2 operator $\begin{pmatrix}0&A^T\\A&0\end{pmatrix}$ with $A=\frac{d}{d\rho}+W$. Boundary conditions (bag): $u(\rho_L)=0$, $v(\rho_R)=0$ for $\chi=(u,v)$; the bag operator is $T^0=-i\alpha\beta$. The maps to the tortoise coordinate are

primordial: $x_0=\dfrac{\arccos e^{-6H\rho}}{6H}$, $B=1$; modern: $x_0=r_s\left(1+W_0\left(e^{\rho/r_s-1}\right)\right)$, $B=\sqrt{1-r_s/x_0}$; future: $\rho=\displaystyle\int_3^{x_0}\frac{dy}{f(y)}$

where W_0 is the Lambert function. **Numerical method.** Staggered grid (u on integer, v on half-integer points, spacing $d=0.02$): A becomes a lower bidiagonal matrix with diagonal $\frac{1}{d}+\frac{W_j}{2}$ and sub-diagonal $-\frac{1}{d}+\frac{W_j}{2}$, and the eigenvalues of h_{KS} are exactly $\pm s_k$, the singular values of A. Test against the exact bag spectrum $k\cos kL+m\sin kL=0$, $\epsilon=\sqrt{k^2+m^2}$ ($m=1$, $L=5$): numerical $1.13204, 1.47963, 1.95318$ versus exact $1.13211, 1.47986, 1.95361$.

Occupations and the two vacua. Because $\sigma_{16}=K\beta$, the $K=-1$ copies enter $\bar\psi\psi$ with a minus sign: $\bar\psi\psi=S_+-S_-$. In terms of one copy's sea density ρ_{copy}:

$$\rho_s=4\rho_{copy}+4\zeta\rho_{copy}+(\text{excitation}),\qquad\zeta=+1\ (\text{vacuum }V_1),\qquad\zeta=-1\ (\text{vacuum }V_2).$$

V_1 (Spin(4,4)-symmetric vacuum): both sectors fill their negative-field-energy levels; the condensates add and a dynamical mass is generated (Nambu-Jona-Lasinio-like, the grid spacing being the ultraviolet cutoff). V_2 (zero-energy mirror vacuum): both sectors fill $\epsilon<0$; the condensates cancel exactly, $\rho_s=0$, $W=Bm$, and the total vacuum energy is exactly zero. **Total energy:** $E=\sum_{occ}(\text{field energy of level})-\frac{\lambda}{2}\int B\rho_s^2\,d\rho$, where a level ϵ of the $K=\pm1$ sector carries field energy $\pm\epsilon$. **First excited state:** ΔSCF, i.e. one particle-hole pair in the lowest level of one copy, re-solved self-consistently.

Thermodynamic effective potential. At temperature T (chemical potential 0) the Mermin-Kohn-Sham grand potential of the fermion gas, solved self-consistently, is

$$\Omega(T)=8\sum_k\left(-s_k-2T\ln\left(1+e^{-s_k/T}\right)\right)-\frac{\lambda}{2}\int B\rho_s^2\,d\rho,\qquad\rho_s=-8\sum_k\tanh\frac{s_k}{2T}\,\rho_k(\rho).$$

5. Results (parameters $H = r_s = m = 1$, $H_\Lambda = 0.1$, $\lambda = -0.03$, $d = 0.02$)

quantity	primordial	modern	future
tortoise box, x_0 range	[0,4]; 0.0572-0.2618	[-4,6]; 1.0068-4.6856	[-3,3]; 1.5599-4.7480
V2: $\max\|W/B - m\|$	0	0	0
V2: vacuum energy	0	0	0
V2: lowest KS levels $\pm s_k$	1.18792, 1.66827, 2.29936	0.478383, 0.833191, 1.04954	0.859668, 1.15294, 1.56262
V2: first excited, + universe pair	+2.35992	+0.954610	+1.71130
V2: first excited, - universe pair	-2.39177	-0.958925	-1.72737
V2: both created together	0 (exact)	0 (exact)	0 (exact)
V1: self-consistent M range	1.5353-6.2331	1.1002-6.1046	1.3499-6.1744
V1: ground energy E_0	-101780.4385	-254547.9728	-152691.6224
V1: $E_1 - E_0$ (ΔSCF)	3.33886	1.03858	2.23828
V1: lowest KS levels $\pm s_k$	1.68180, 2.07587, 2.62805	0.520477, 0.971297, 1.24416	1.12479, 1.37113, 1.73680

Thermodynamic effective potential $\Omega(T)$ and mean effective mass (vacuum V_1); $T_H = 1/(4\pi r_s) = 0.0795775$ is the Hawking temperature of the modern bridge:

T	Ω primordial	$\bar{M}$	Ω modern	$\bar{M}$	Ω future	$\bar{M}$
0.0795775	-101780.4385	1.60220	-254547.9746	1.32224	-152691.6224	1.43383
0.25	-101780.4444	1.60198	-254548.5722	1.31644	-152691.6880	1.43236
0.5	-101780.8961	1.59423	-254553.1705	1.29929	-152693.3587	1.41643
1	-101787.4992	1.54950	-254579.3755	1.26456	-152707.2232	1.37102
2	-101829.6777	1.46201	-254699.1228	1.21945	-152776.9684	1.30837

The effective mass decreases with temperature (thermal melting of the condensate), as expected for a Kohn-Sham fermion gas with an attractive ($\lambda < 0$) scalar coupling.

6. Theorems: universes of masses {+m, -m} are created in pairs

Definitions. A **Universe** is a mode function of dirac16complex (a c-number solution of the linear or Kohn-Sham field equation), as in Pre-gravity6plane.nb. Its **mass** is its energy ϵ with respect to translations of the time x_4 (for zero momentum, $\epsilon = \pm M$), or, for the mass-sign theorem, the parameter m of the equation it solves.

Theorem 1 (mass-sign pairing; dirac16complex; primordial, modern and future fields). Let $\mathcal{C}\psi = \Gamma\psi^*$ ($\Gamma = T^0 \cdots T^7$) and $\mathcal{P}_1\psi(x) = T^1\psi(R_1 x)$, $R_1 : x_1 \mapsto -x_1$. Then, as exact identities in the Grassmann algebra,

$$\mathcal{L}_{m,\lambda}[\mathcal{C}\psi] = \mathcal{L}_{-m,\lambda}[\psi], \qquad \mathcal{L}_{m,\lambda}[\mathcal{P}_1\psi](x) = \mathcal{L}_{-m,\lambda}[\psi](R_1 x).$$

Proof. Γ commutes with σ_{16} and with every S^{AB}, anticommutes with every T^A, is real with $\Gamma^2 = I$. Under $\psi \to \Gamma\psi^*$ the reordering of Grassmann factors gives $\bar{\psi}T^A\partial_A\psi \to \bar{\psi}T^A\partial_A\psi$ (up to the same antisymmetrized combination), $\bar{\psi}\psi \to -\bar{\psi}\psi$, $(\bar{\psi}\psi)^2 \to (\bar{\psi}\psi)^2$; the vielbein and Ω_μ are real and commute with Γ up to the same signs. For $\mathcal{P}_1$: $(T^1)^T\sigma_{16}T^1 = -\sigma_{16}$ and $T^1T^AT^1 = -T^A$ for $A \neq 1$, $T^1T^1T^1 = T^1$ together with $\partial_1 \to -\partial_1$ leave the kinetic term invariant and flip $\bar{\psi}\psi$; R_1 is an isometry of all three fields. Both identities were verified symbolically in each field (6planes.nb, Part 5.1). $\mathcal{C}^2 = \mathcal{P}_1^2 = 1$, so each map is a bijection between the solution sets with masses $+m$ and $-m$. □

Theorem 2 (spectral pairing; all three fields). For every real effective potential W,

$$T^0 h_{KS} T^0 = -h_{KS}, \qquad h_{KS}^* = -h_{KS},$$

and both maps preserve the bag boundary conditions. Hence every Kohn-Sham level ϵ has a partner $-\epsilon$: a Universe of mass $+\epsilon$ is always accompanied by a Universe of mass $-\epsilon$. *Proof.* $T^0\alpha T^0 = -\alpha$, $T^0\beta T^0 = -\beta$; α is real and β imaginary, and $p^* = -p$ (verified, 6planes.nb Part 6.1); T^0 is the bag operator itself. The full field equation of dirac16complex has real coefficients, so complex conjugation maps a stationary solution of frequency ϵ to one of frequency $-\epsilon$ also for configurations that depend on all eight coordinates. Numerically the spectrum is $\{\pm s_k\}$ by construction (singular values). □

Theorem 3 (mirror sectors and zero-energy pair creation; every configuration of dirac16complex independent of x_5, x_6, x_7).

$$\mathcal{L} = \mathcal{L}_D[\psi_+] - \mathcal{L}_D[\psi_-] - \frac{\lambda}{2}(S_+ - S_-)^2, \qquad (\bar{\psi}\psi)^2 = S_+^2 + S_-^2 - 2S_+S_-, \qquad K\psi_\pm = \pm\psi_\pm.$$

(i) The $\pm$ sectors carry opposite field energies: a $+$ Universe of energy ϵ and a $-$ Universe in the same orbital have energies $+\epsilon$ and $-\epsilon$. (ii) In the zero-energy vacuum V_2 the total energy is exactly 0 and the condensates cancel. (iii) Creating a $+$ Universe pair (energy $+\Delta$) together with a $-$ Universe pair (energy $-\Delta$) leaves ρ_s and the Kohn-Sham field unchanged and changes the total energy by exactly 0 (table above, all three fields): the creation of the pair costs no energy. (iv) The cross term $+\lambda S_+S_-$ (232 Grassmann monomials) connects the vacuum with this state, so the process is allowed by the dynamics; the separately conserved charges $Q_\pm = \int \psi_\pm^\dagger \psi_\pm$ forbid the creation of a single quantum, and conservation of the total energy (which is 0 in V_2) forbids the creation of quanta of only one sign, so the minimal allowed creation is the $\{+,-\}$ pair. Therefore **universes of masses $\{+m, -m\}$ are created in pairs**, in each of the primordial, modern and future fields. *Proof.* Algebra of Part 3.6 and Part 7 of 6planes.nb (exact identities) and the Kohn-Sham energy balance, in which the field-energy contributions $+2s_1$ and $-2s_1$ cancel and ρ_s is unchanged. □

Remarks (honest scope). The theorems are exact statements about the model defined here: the Lagrangian of Section 1, the three gravitational fields of Section 2, and the Kohn-Sham approximation of Section 4. They do not prove that the physical universe was created this way. The single $\pm$ excitations are not exact mirror images of each other ($+2.35992$ versus -2.39177 in the primordial field), because the self-interaction energy $-\frac{\lambda}{2}\int B\rho_s^2$ is even under the exchange of the sectors. The exact cancellation holds for the pair. With four time directions the quantized field has an indefinite (Krein) metric; the $-$ sector consists of negative-norm (equivalently negative-energy) quanta.